Calculus Practice — Optimization with Derivatives

8 questions. Try all of them before checking the answers below.

Practice Questions

Q1 Find the minimum of $f(x) = x^2 + \frac{16}{x}$ for $x > 0$, and the value of x at which it occurs.

Q2 A power line needs to connect to three houses located at positions $x = 1$, $x = 4$, and $x = 9$ along a straight road. Find the position x that minimizes the total squared distance:

$$D(x) = (x-1)^2 + (x-4)^2 + (x-9)^2$$

Q3 The softplus function is defined as $L(z) = \ln(1 + e^z)$. Find $L'(z)$, and simplify your answer in terms of the sigmoid function $\sigma(z) = \frac{1}{1+e^{-z}}$.

Q4 A simple one-parameter linear model $y = mx$ is fit to the data points $(1, 2), (2, 4), (3, 5)$ by minimizing the squared-error cost function:

$$C(m) = (2-m)^2 + (4-2m)^2 + (5-3m)^2$$

Find the value of m that minimizes $C(m)$.

Q5 Find the critical points of $f(x) = x^3 - 6x^2 + 9x + 2$, and classify each as a local maximum or local minimum.

Q6 For a single training example with true label $y = 1$ and predicted probability $p = \sigma(z)$, the log-loss is:

$$L(z) = -\ln\left(\sigma(z)\right)$$

Find $\frac{dL}{dz}$, and show that it simplifies to $\sigma(z) - 1$.

Q7 Find the absolute minimum of $f(x) = 2x^2 - 8x + 3$ and the x-value at which it occurs.

Q8 Find the critical points of $f(x) = x^4 - 4x^3 + 4x^2$, and classify each as a local maximum or local minimum.

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Answers & Solutions

A1. $f'(x) = 2x - \frac{16}{x^2}$. Setting $f'(x) = 0$: $2x^3 = 16 \Rightarrow x^3 = 8 \Rightarrow x = 2$. Since $f''(x) = 2 + \frac{32}{x^3}$, $f''(2) = 6 > 0$, so this is a minimum.

$$f(2) = 4 + 8 = 12$$

A2. $D'(x) = 2(x-1) + 2(x-4) + 2(x-9) = 6x - 28$. Setting $D'(x) = 0$:

$$x = \frac{28}{6} = \frac{14}{3}$$

This is simply the average of $1, 4, 9$. Since $D''(x) = 6 > 0$, this is a minimum, with minimum value $D(\frac{14}{3}) = \frac{98}{3}$.

A3.

$$L'(z) = \frac{e^z}{1 + e^z}$$

Multiplying numerator and denominator by e^{-z}:

$$L'(z) = \frac{1}{1 + e^{-z}} = \sigma(z)$$

So the derivative of the softplus function is just the sigmoid function.

A4. Expanding each term:

$$C(m) = 14m^2 - 50m + 45$$

$$C'(m) = 28m - 50 = 0 \Rightarrow m = \frac{50}{28} = \frac{25}{14}$$

Since $C''(m) = 28 > 0$, this is a minimum.

A5. $f'(x) = 3x^2 - 12x + 9 = 3(x-1)(x-3)$, so the critical points are $x = 1$ and $x = 3$. Using $f''(x) = 6x - 12$: - At $x = 1$: $f''(1) = -6 < 0$ — local maximum, with $f(1) = 6$. - At $x = 3$: $f''(3) = 6 > 0$ — local minimum, with $f(3) = 2$.

A6. Using the chain rule and $\sigma'(z) = \sigma(z)(1 - \sigma(z))$ from before:

$$L'(z) = -\frac{\sigma'(z)}{\sigma(z)} = -\frac{\sigma(z)(1 - \sigma(z))}{\sigma(z)} = -(1 - \sigma(z)) = \sigma(z) - 1$$

A7. $f'(x) = 4x - 8 = 0 \Rightarrow x = 2$. Since $f''(x) = 4 > 0$, this is a minimum.

$$f(2) = 8 - 16 + 3 = -5$$

A8. Notice $f(x) = x^2(x-2)^2$. Then $f'(x) = 4x^3 - 12x^2 + 8x = 4x(x-1)(x-2)$, giving critical points $x = 0, 1, 2$. Using $f''(x) = 12x^2 - 24x + 8$: - At $x = 0$: $f''(0) = 8 > 0$ — local minimum, $f(0) = 0$. - At $x = 1$: $f''(1) = -4 < 0$ — local maximum, $f(1) = 1$. - At $x = 2$: $f''(2) = 8 > 0$ — local minimum, $f(2) = 0$.